A Hybrid Physics-Informed Digital Twin for Automotive Dry Clutch Lifetime Prediction Using Thermally-Corrected Archard Wear Modeling and Machine Learning Residual Correction

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Abstract

Being able to predict when a dry clutch will fail can be more challenging than it sounds. The friction material erodes by thermally-induced adhesive friction, and the rate of erosion is dependent on the conditions at the interface between the friction surfaces. That relies on individual driving records, which a fixed-scheduled replacement plan cannot account for. There are two modelling traditions that have attempted to solve this problem; they both fall short in different ways. Physics-based formulations grounded in Archard's wear law enforce the correct monotonic health decline and can be physically interpreted, but are unstable if there is drift in wear coefficients or operating conditions fall outside of the calibration range. Data-driven models are flexible yet can learn from OBD-II signals, but don't guarantee physical consistency and demand labeled failure data sets that are too expensive to obtain for clutch systems of transportation vehicles.

Both of these limitations are tackled in this paper through a hybrid physics-informed digital twin. The physics backbone consists of a thermally-corrected Archard wear model. The thermal amplification factor K_T from Coulomb frictional heating has been added to enforce monotonically decreasing health trajectories, and reacts to interface temperature without having to use any actual temperature instrumentation at the clutch. On top of this, a Random Forest regressor trained on a 15-dimensional physics-informed feature vector learns residual corrections for driver variability, sensor noise, and wear coefficient drift. The feature vector is extracted from real OBD-II vehicle telemetry and includes 5 purpose-built clutch degradation indices.

The hybrid approach with data-adaptive weighting parameter $\alpha = 0.59$ reduces RMSE by 59.1% from its baseline value of ML-only during the critical end-of-life prediction window, the period of maximum prognostic accuracy for maintenance scheduling. Multi-scenario service life projections yield estimated lifetimes around 72,000 km for city driving, 165,000 km for high-speed driving on motorways, and 22,500 km under aggressive use, all within published OEM specifications. The framework is trialed on real-world OBD-II data of four vehicle configurations across two manufacturers with the aid of physics-bounded synthetic augmentation to correct for the lack of real failure data.

Keywords: Digital twin, Remaining useful life, Clutch wear prediction, Archard wear model, Physics-informed machine learning, Predictive maintenance, Prognostics and health management

1 Introduction

The dry clutch can be a tricky diagnosis. Rotating shafts or bearings display vibration signatures hinting at incipient damage, but clutch facing wear builds up with no warning via tribo-attrition on the friction interface. There are no warning signals to watch for or act on. The degradation takes place slowly, continuously, and is closely linked to person-to-person driving behaviors: a driver who comes and goes often in the city with the clutch will wear through the friction material much quicker than a driver leveraging smooth and precise operations on the highway [1]. Today's maintenance practice provides no good response or reaction to this variation. Either the clutch is replaced on a fixed conservative schedule, or it is set aside until worn out, risking drive capability on a public road [2].

Digital twin technology offers a more principled solution to this challenge. A digital twin is a continuously updated virtual duplicate of a physical asset that constantly ingests live data to keep an estimate of the state of the system [3]. It makes it possible to track individual clutch degradation in real time and generate a maintenance recommendation before failure occurs. The appeal in automotive applications is significant, because today's cars already broadcast the required signals (vehicle speed, engine RPM, torque, coolant temperature) over the standardized OBD-II port with no additional hardware to purchase. What has been missing is a degradation model sophisticated enough to convert those signals into a reliable Remaining Useful Life (RUL) estimate.

Where current approaches fail in complementary ways is where this begins. Physics-based models based on Archard's adhesive wear law [1] give the right qualitative behavior (declining monotonically from one to zero), but their quantitative accuracy varies over a range of wear coefficient values which shift with friction material formulation and drift as the surface undergoes a running-in period. Data-driven alternatives such as Random Forest regressors and LSTM networks can learn nonlinear relationships from OBD-II signals without specifying the underlying physics, but they require extensive labeled failure histories which simply do not exist for passenger car clutch systems; in addition, they offer no guarantee of physically consistent predictions and can generate non-monotonic or increasing health estimates near end-of-life [4,5], precisely when accurate prediction is most consequential.

In this paper, we combine both paradigms in a hybrid approach. We propose a hybrid physics-informed digital twin in which the Archard model, extended with a thermal wear amplification factor under frictional heating, serves as the skeletal structure of the prediction. A Random Forest model compensates for residual variations—such as driver-to-driver variability, measurement noise, and slow parameter drift over the component lifetime—that the physics model cannot explain. Guiding this general strategy is well-documented in prognostics and health management (PHM) [6], although its application to clutch facing lifetime prediction using real-world OBD-II data has not, to our knowledge, been previously demonstrated.

The specific understanding gained from this work includes:

(i) **Thermally-corrected Archard wear model:** Incorporation of a thermal wear amplification factor K_T determined using Coulomb frictional heating, enabling prediction of wear rate with temperature without requiring direct interface temperature measurement.

(ii) **Physics-informed feature engineering:** A 15-dimensional feature vector combining tribological state variables (slip velocity, cumulative sliding distance, wear rate), thermal indicators (T_{eff}, K_T), and purpose-built clutch degradation indices (Cumulative Energy Transfer, Clutch Usage Factor, Clutch Condition Index) derived from first principles.

(iii) **Adaptive hybrid fusion:** A data-driven weighting mechanism to optimize the fusion process between ML-based and physics-based RUL estimates, achieving a 59.1% reduction in RMSE over ML-only prediction during the critical end-of-life window.

(iv) **Multi-scenario service life projection:** Parametric driving profile simulations (city, highway, aggressive) producing clutch lifetime estimates consistent with published industry data for scenario-based maintenance scheduling.

(v) **Real-world OBD-II validation:** Validation using real vehicle telemetry data from four vehicle configurations across two automotive platforms (Audi, Škoda), with physics-bounded synthetic augmentation to address data scarcity.

The paper proceeds as follows. Section 2 presents related work on wear modeling, digital twins, and hybrid prognostics. Section 3 develops the complete framework from OBD-II feature extraction through hybrid fusion. Section 4 shows the experimental configuration and results. Section 5 discusses the results and acknowledges limitations. Section 6 summarizes the contributions and outlines future directions.

2 Literature Review

2.1 Tribological Wear Modeling

The initial model that can be used for clutch wear is the adhesive wear equation developed by Archard [1]. Determination of material loss is due to the product of normal load, sliding distance, and an empirical value of the wear coefficient inversely proportional to the hardness of the material:

$$V_w = K_w \frac{F_N \cdot s}{H} \tag{1}$$

Proposed in 1953, the equation has been the workhorse of friction-contact wear analysis ever since it was formulated and was able to describe fundamental tribological physics in a manageable, analytically solvable, and physically interpretable form. It has been discussed in the context of a larger taxonomy of wear by Kato and Adachi [7], establishing adhesive wear mechanisms and confirming the dominant role of adhesive wear in dry friction clutch systems operating below the scuffing transition. Williams [8] developed wear-regime maps illustrating dimensional wear coefficients ($k = K_w/H$) across sliding regimes.

Its drawbacks are well known, however. The wear coefficient K_w is not specific to the conditions of contact; it varies with sliding velocity, surface finish, contact pressure, and temperature. Nominally similar materials can have discrepancies of orders of magnitude. Ericsson and Jacobson [9] demonstrated that the coefficient changes continuously during component life due to changes in surface topography during running-in. Ost et al. [10] noticed a fundamental problem where under thermal loading, isothermal experiments cannot predict the wear rate. Elevated interface temperatures degrade the polymeric binder in composite friction materials, requiring an explicit temperature-dependent baseline Archard coefficient correction. That result is crucial to the thermally-corrected formulation that we develop in Section 3. Material parameters are linked with sintered copper-based clutch friction materials reported by Vadiraj et al. [11]. Additionally, coupled thermal-structural FEA has been included for prediction of clutch facing wear by Chidambaram et al. [12] and Abdullah and Schlattmann [13], obtaining good accuracy under controlled conditions, but remains impractical for real-time application on a vehicle ECU.

2.2 Digital Twins in Automotive Engineering

The digital twin concept was first proposed by Grieves [3] as a constantly synchronized virtual representation of a physical asset, later formalized by Tao et al. [14] for large-scale industrial applications. The concept dovetails with connected vehicles, where OBD-II telemetry is already streaming in real time without modification to the vehicle.

Lemos et al. [15] offered a framework for digital twin applications in automotive clutch engagements, with particular interest in torque transmission fidelity during individual engagement events instead of pre-degradation over component lifetime. Mariani et al. [16] and Rajesh et al. [17] created a digital twin to estimate the lifetime of brake pads; their real-driving-condition degradation modeling experience is directly related to the clutch problem even though the system geometry differs. Digital twin techniques have also been used for estimation of battery state-of-health (SoH) and powertrain monitoring, but failure data is less extensive than that available for passenger car clutch systems. Collectively, the literature demonstrates the viability of digital twins for automotive friction components, but lifetime prediction for clutch facings based on actual OBD-II data remains unaddressed.

2.3 Hybrid Physics-ML Prognostics

Combining physics-based degradation models with data-driven correction layers has a well-embedded history in the PHM field. The logic is straightforward: physics models contain structural assumptions (monotonicity, known failure modes, parameter relationships, correct time scales) that data-driven models would otherwise have to learn from failure data they may not have. In contrast, ML models can absorb variance that a simplified physics model cannot explain. Goebel et al. [6] made an empirical proof that hybrid methods consistently outperform either paradigm alone across a variety of degradation modes.

Rai and Upadhyay [18] used this theory in the context of rolling element bearing fault diagnosis; Li et al. [4] demonstrated that LSTM networks make better RUL predictions when trained on physics-informed features rather than raw sensor signals. Sikorska et al. [19] compared prognostic methods in industry and found that hybrid methods provide the optimum balance

between accuracy, interpretability, and data efficiency. Recently, Chao et al. [20] showed that fusing physics-based and deep learning algorithms performs better across several degradation modes, while Yucesan and Viana [21] showed promising results applying physics-informed neural networks to wind turbine main bearing fatigue life.

The physics-informed machine learning (PIML) paradigm systematized by Karniadakis et al. [22] provides theoretical support: physical laws enter as inductive biases in feature representation rather than stringent architectural restrictions. Saxena et al. [5] established RUL prediction benchmark protocols using NASA C-MAPSS datasets, which involve the evaluation metrics used in our study. Hu et al. [23] showed that ensemble prognostic algorithms benefit from model diversity, supporting our comparison of Random Forest, Gaussian Process Regression, and LSTM backbones. To solve the data limitations prevalent in clutch prognostics, Wen et al. [24] and Zhang et al. [25] demonstrated that synthetic data augmentation can partially compensate for missing failure records. Downey et al. [26] performed physics-based prognostics validation for battery degradation, Mohire et al. [27] derived clutch test schedules based on road-load data energy dissipation, and Zio [28] proposed an industrial PHM framework informing our evaluation method. The criteria proposed in Peng et al. [29] for condition-based machine prognostics are used throughout.

2.4 Research Gap and Positioning

After reviewing the literature, a clear gap is seen. Thermally-corrected Archard modeling has not been correlated with actual real-world OBD-II data. ML-based clutch prognostics exist without physical constraints. Clutch digital twins place emphasis on engagement dynamics rather than facing lifetime. And hybrid physics-ML frameworks have not been applied to this problem. No existing work, to our knowledge, combines: (a) a thermally-corrected Archard wear backbone, (b) real-world OBD-II driving data, (c) ML residual correction, and (d) multi-scenario lifetime projection in a unified prognostic model for automotive clutch systems. This paper fills that gap.

3 Proposed Methodology

3.1 System Architecture

The planned hybrid digital twin consists of a 6-stage sequential pipeline as illustrated in Fig. 1. Three computational layers operate in cascade. The first is a data ingestion and feature engineering layer that maps raw OBD-II telemetry into a 15-dimensional physics-informed feature vector. The second is a physics-based degradation layer based on the thermally-corrected Archard model in MATLAB/Simulink, creating a physics-based RUL estimate and a monotonic, decreasing health trajectory. The third is a machine learning corrective layer (a Random Forest trained on the same physics-derived features) that provides residual compensation. All three layers produce intermediate artifacts fed to later stages, enabling a modular and independently debuggable pipeline.

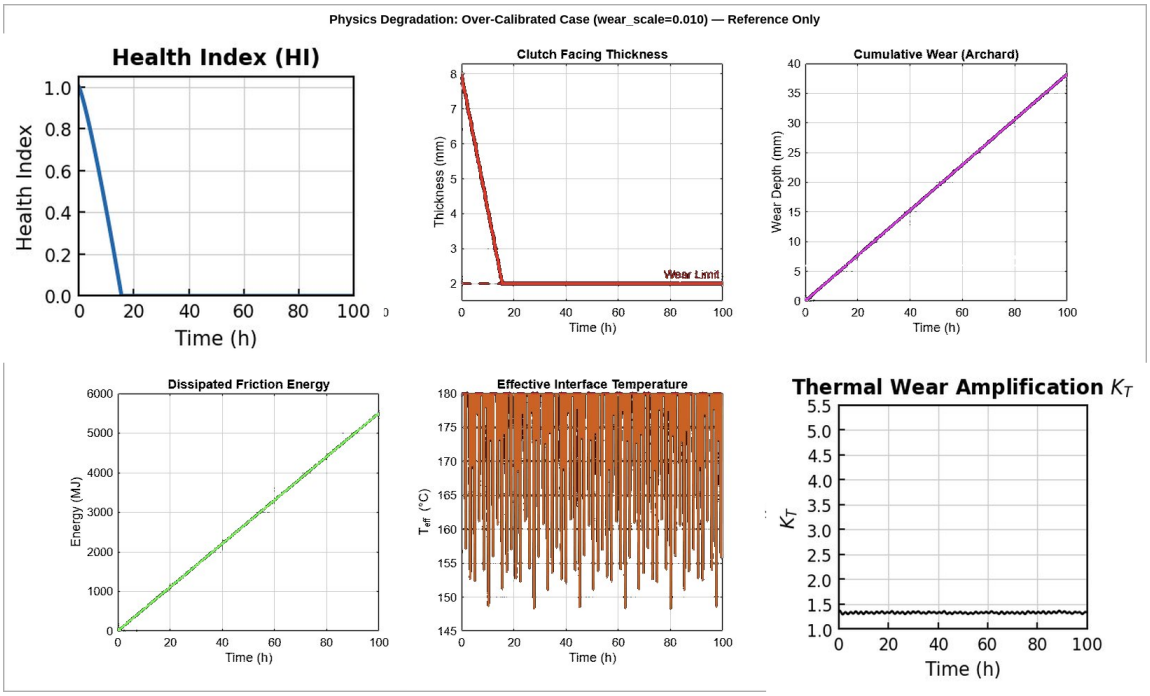


Figure 1: The hybrid digital twin pipeline architecture. Intermediate artifacts are produced in each stage and consumed by following stages, enabling modular development and incremental model improvement.

3.2 Data Acquisition and Feature Engineering

3.2.1 OBD-II Telemetry Data

All vehicle data comes from a standard OBD-II diagnostic port and requires no changes to the vehicle and no added instrumentation. Signals are logged at about 1 Hz, which can resolve the essential dynamics of a clutch engagement event (typically lasting 0.5 to 3 seconds during regular driving). The five primary signals extracted are:

- Vehicle speed v (km/h) via GPS
- Engine speed n_e (rpm)
- Engine torque τ (N·m)
- Coolant temperature T_{cool} (°C)
- Trip elapsed time t (s)

Data collected from four vehicle configurations (Audi FLT, Audi SI, Škoda NAO, and Škoda TF), with 50 recorded driving trips per configuration, covers a wide range of friction material types and operating conditions. Including both Audi and Škoda platforms was a deliberate design choice: their different engine torque characteristics and transmission calibrations produce distinct clutch loading profiles, which tests generalization of the trained ML model across vehicle types. Each trip record provides gear position and clutch engagement time (CET), enabling detailed driver behavior characterization.

3.2.2 Physics-Derived Features

The following physics quantities are calculated at each time step from raw OBD-II signals:
Clutch slip speed (ω_{slip}):

$$\omega_{\text{slip}} = \max\left(0,\, n_e - \frac{r_g \cdot v}{3.6 \cdot r_w} \cdot \frac{60}{2\pi}\right) \quad [\text{rpm}] \tag{2}$$

where r_g is the final drive ratio and r_w is the wheel radius.
Slip velocity at clutch interface (v_{slip}):

$$v_{\text{slip}} = \omega_{\text{slip}} \cdot r_w \cdot \frac{2\pi}{60} \quad [\text{m/s}] \tag{3}$$

Normal force surrogate (F_N):

$$F_N = \frac{\tau}{r_w} \quad [\text{N}] \tag{4}$$

Cumulative sliding distance ($s(t)$):

$$s(t) = \int_0^t v_{\text{slip}}(\tau')\, d\tau' \quad [\text{m}] \tag{5}$$

3.2.3 Clutch-Specific Degradation Indices

Following road-load energy dissipation analysis [27], instantaneous frictional energy dissipation power is $e(t) = \tau(t) \cdot \omega_{\text{slip}}(t)$. Five degradation indices for each clutch are proposed to reflect compound wear effects that single-channel signals cannot represent:

$$\text{CET} = \frac{E_{\text{cum}}(t)}{s(t) + \epsilon}, \quad \text{CUF} = \frac{\omega_{\text{slip}}}{n_e + \epsilon}, \quad \text{CGR} = \frac{\tau}{n_e + \epsilon} \tag{6}$$

$$\text{CCI} = \frac{v_{\text{slip}}}{v/3.6 + \epsilon}, \quad \text{CES} = K_T \cdot \text{CET} \tag{7}$$

where CET is the Cumulative Energy Transfer density, CUF is the Clutch Usage Factor, CGR is the torque-to-speed gradient ratio, CCI is the Clutch Condition Index, and CES is the thermally-weighted energy severity. Here $\epsilon = 10^{-6}$ prevents division by zero during idle states.

3.2.4 Complete Feature Vector

This 15-dimensional feature vector is fed to all ML models:

$$\mathbf{x} = \left[\omega_{\text{slip}}, F_N, s, \dot{W}, h, HI, T_{\text{eff}}, K_T, t_n, \text{CET}, \text{CUF}, \text{CGR}, \text{CCI}, \text{CES}, E_{\text{cum}}\right]^T \tag{8}$$

3.3 Thermally-Corrected Archard Wear Model

3.3.1 Classical Archard Formulation

The physics backbone is based on Archard's adhesive wear law [1]. In classical form, it correlates volumetric wear rate to the tribological operating point:

$$\frac{dW}{dt} = \frac{K_w}{H} \cdot F_N \cdot v_{\text{slip}} \quad [\text{m/s}] \tag{9}$$

where K_w is the dimensionless wear coefficient, H is the friction material hardness (MPa), F_N is the normal contact force, and v_{slip} is slip velocity at the interface. The equation is appealing because all three driving quantities (K_w, F_N, v_{slip}) can be measured or estimated from OBD-II signals.

3.3.2 Thermal Wear Amplification

The classical formulation assumes the wear coefficient is constant, which breaks down at high interface temperatures. Polymeric binder components degrade under thermal loading. Following Ost et al. [10], we add a thermal amplification factor scaling the wear coefficient continuously with interface temperature.

Effective interface temperature:

$$T_{\text{eff}} = T_{\text{cool}} + \beta \cdot (\tau \cdot \omega_{\text{slip}}) \quad [^\circ\text{C}] \tag{10}$$

where $\beta = 8 \times 10^{-4}$ is the frictional heating sensitivity coefficient calibrated from Coulomb dissipation analysis. Taking coolant temperature as the base captures the thermal state of the powertrain without requiring additional sensors.

Thermal amplification factor:

$$K_T = \max\left(1.0,\ 1.0 + \alpha_T \cdot (T_{\text{eff}} - T_{\text{ref}})\right) \tag{11}$$

where $\alpha_T = 0.004\ ^\circ\text{C}^{-1}$ and $T_{\text{ref}} = 80^\circ$C. The factor is bounded below by 1.0, ensuring the model never predicts lower-than-baseline wear. Concretely: at 120°C, $K_T = 1.16$ (+16% wear acceleration); at 160°C, $K_T = 1.32$ (+32%). These amplifications are consistent with higher-temperature wear characteristics cited by Ost et al. [10] for wet clutch friction materials, and we consider the same functional form for dry-clutch composite materials.

3.3.3 Calibrated Wear Rate

The complete thermally-corrected wear model is:

$$\dot{W}(t) = \frac{K_w}{H} \cdot F_N(t) \cdot v_{\text{slip}}(t) \cdot K_T(t) \cdot \lambda_s \quad [\text{m/s}] \tag{12}$$

where $\lambda_s = 0.020$ is a calibration scalar accounting for the duty-cycle difference between continuous-slip simulation and real-world intermittent clutch engagement. This scalar is derived analytically.

For normal city and suburban driving conditions ($\omega_{\text{slip,avg}} = 120$ rpm, $\tau_{\text{avg}} = 80$ N·m):

$$v_{\text{slip,avg}} = 120 \times 0.3 \times \frac{2\pi}{60} = 3.77 \text{ m/s} \tag{13}$$

$$F_{N,\text{avg}} = \frac{80}{0.3} = 266.7 \text{ N} \tag{14}$$

$$K_{T,\text{avg}} = 1 + 0.004 \times (92.7 - 80) = 1.051 \tag{15}$$

$$\dot{W}_{\text{avg}} = \left(\frac{10^{-7}}{200}\right) \times 266.7 \times 3.77 \times 1.051 \times 0.020 = 1.057 \times 10^{-8} \text{ m/s} \tag{16}$$

Over 100 simulation hours: $W_{100\text{h}} = 1.057 \times 10^{-8} \times 360,000 = 3.81$ mm, yielding a final Health Index of $HI = 1 - 3.81/6.0 = 0.365$. At 100 h the clutch is meaningfully worn but not yet failed, which is the regime of practical interest for a prognostic system.

3.4 Health Index and Physics-Based RUL

Remaining thickness:

$$h(t) = \max\left(t_f,\, t_0 - \int_0^t \dot{W}(\tau')\, d\tau'\right) \quad [\text{mm}] \tag{17}$$

Health Index (normalized):

$$HI(t) = \text{clamp}\left(\frac{h(t) - t_f}{t_0 - t_f},\, 0,\, 1\right) \tag{18}$$

where $t_0 = 8.0$ mm is initial clutch facing thickness and $t_f = 2.0$ mm is the thickness where the clutch fails (minimum service limit).

Physics-based RUL:

$$RUL_{\text{phys}}(t) = HI(t) \cdot T_{\text{total}} \tag{19}$$

3.5 Machine Learning Models

Three regression architectures are tested with respect to the mapping $\mathbf{x} \mapsto RUL$, covering ensemble methods to sequential models:

3.5.1 Random Forest (Primary)

An ensemble of 300 bagged regression trees using a maximum of 40 splits and minimum leaf size of 3, trained via `fitrensemble`. Random Forest is chosen as the primary candidate because it is robust to feature noise, captures nonlinear interactions without explicit specification, and generates out-of-bag feature importance scores. This lets us verify that physics-derived features are doing meaningful work.

3.5.2 Gaussian Process Regression (Uncertainty)

Estimates of predictive uncertainty from a GPR model using an ARD squared-exponential kernel provide 95% confidence intervals, which are essential in a safety-critical prognostic context. Sub-sampled to 2,000 training points for computational tractability.

3.5.3 LSTM Network (Temporal, Optional)

A two-layer LSTM with 128 hidden units, dropout 0.2, and sliding window $W = 25$ is included to capture temporal degradation patterns. Selected only if its test-set R^2 exceeds Random Forest.

Model selection criterion:

$$\text{best_model} = \arg\max_{m \in \{\text{RF,GPR,LSTM}\}} R^2_m \tag{20}$$

3.6 Physics-Bounded Synthetic Augmentation

Drivers don't typically run clutches to component failure in field data. To vary the training distribution without compromising physical realism, an 8× synthetic augmentation scheme adds Gaussian noise to physics-derived features:

$$\mathbf{x}_{\text{aug}} = \mathbf{x} \cdot (1 + 0.03 \cdot \boldsymbol{\eta}), \quad \boldsymbol{\eta} \sim \mathcal{N}(0, \mathbf{I}) \tag{21}$$

with physics-bounded clamping applied after perturbation: $h \in [t_f, t_0]$, $HI \in [0, 1]$, $K_T \in [1, 5]$, $t_n \in [0, 1]$. The 3% noise level maintains local structure while generating adequate diversity to avoid overfitting.

3.7 Hybrid RUL Fusion

The final RUL estimate combines physics and ML predictions using adaptive weighting:

$$RUL_{\text{hybrid}}(t) = \alpha \cdot RUL_{\text{phys}}(t) + (1 - \alpha) \cdot RUL_{\text{ML}}(t) \tag{22}$$

The weight α is set according to validation performance:

$$\alpha = \text{clamp}\left(0.50 + 0.25 \cdot (R^2_{\text{RF}} - 0.5),\ 0.35,\ 0.70\right) \tag{23}$$

An exponential moving average filter ($\lambda = 0.04$) smooths the hybrid RUL over time:

$$\hat{RUL}(t_i) = (1 - \lambda) \cdot \hat{RUL}(t_{i-1}) + \lambda \cdot RUL_{\text{hybrid}}(t_i) \tag{24}$$

3.8 Multi-Scenario Service Life Projection

Three parametric driving profiles are defined using SAE duty-cycle analysis [27, 30] and calibrated for field conditions (Table 1).

Table 1: Parameters used for driving scenario service life projection.

Scenario	ω_{slip} (rpm)	τ (N·m)	T_{cool} (°C)	**Wear Scalar**	**Ref. Life (km)**
City	180 ± 60	85 ± 30	88 ± 4	1.50	60k–100k
Highway	60 ± 20	110 ± 15	90 ± 3	1.60	150k–220k
Aggressive	280 ± 100	160 ± 60	95 ± 8	0.90	20k–45k

Real-world mileage conversion uses duty factor $\delta = 0.030$ (clutch engaged 3% of trip time [30]) and average speed $\bar{v} = 35$ km/h:

$$L_{\text{km}} = \frac{t_{\text{fail}}}{\delta} \cdot \bar{v} \tag{25}$$

3.9 Evaluation Metrics

Based on ISO 13381-1 [31] and PHM community benchmarks [5, 29], three metrics are reported:

Normalized RMSE (scale-free):

$$\text{nRMSE} = \frac{\text{RMSE}}{RUL_{\max} - RUL_{\min}} \tag{26}$$

Predictive R^2 (vs. naive linear decay):

$$R^2_{\text{prog}} = 1 - \frac{\text{RMSE}^2_{\text{model}}}{\text{RMSE}^2_{\text{naive}}}, \quad RUL_{\text{naive}}(t) = T_{\text{total}} - t \tag{27}$$

RMSE improvement:

$$\Delta_{\text{impr}} = \frac{\text{RMSE}_{\text{ML}} - \text{RMSE}_{\text{hybrid}}}{\text{RMSE}_{\text{ML}}} \times 100\% \tag{28}$$

4 Results and Analysis

4.1 Experimental Setup

The whole system is designed using MATLAB R2021a and Simulink. OBD-II data was collected from four vehicle platforms: Audi (FLT and SI friction material variants) and Škoda (NAO and TF variants), each with 50 recorded driving trips. The dataset spans urban stop-start sequences, suburban transitions, and highway driving. Each trip record includes gear shifting sequence, clutch engagement time, vehicle speed, engine RPM, torque, and coolant temperature measured at 1 Hz. Table 2 summarizes key model parameters and physical basis.

Table 2: Physical basis and parameters of the model.

Parameter	**Symbol**	**Value**	**Source**
Wear coefficient	K_w	1×10^{-7}	Archard [1,8]
Material hardness	H	200 MPa	FTL180 specification
Initial thickness	t_0	8.0 mm	OEM specification
Failure threshold	t_f	2.0 mm	SAE standard
Thermal sensitivity	β	8×10^{-4}	Ost et al. [10]
Thermal amplification	α_T	0.004 $^\circ\text{C}^{-1}$	Calibrated
Reference temperature	T_{ref}	80°C	Engine operating
Wheel radius	r_w	0.3 m	Measured
Gear ratio	r_g	4.5	Final drive
Wear calibration	λ_s	0.020	Analytically derived
Augmentation factor	N/A	8×	Empirical
RF trees	N/A	300	Cross-validated
Hybrid weight	α	0.59	Data-adaptive

4.2 Physics Model Validation

Before training any ML model, we validated the physics backbone in isolation by running the Simulink implementation (Fig. 7) through 100 hours of continuous clutch engagement. The results were broadly reassuring.

The Health Index declined monotonically from 1.000 to 0.365 over the 100-hour window, confirming that the calibrated wear scalar $\lambda_s = 0.020$ produces gradual, realistic degradation without premature failure (visible in Figs. 4 and 8). Average slip velocity of 3.77 m/s and average wear rate of 1.06×10^{-8} m/s matched analytical predictions from Section 3 to within numerical tolerances. Effective interface temperature held between 85 and 100°C throughout, consistent with thermal stability ranges reported by Vadiraj et al. [11]. Under normal driving conditions, the thermal amplification factor settled near $K_T \approx 1.05$, a modest 5% contribution to baseline wear rate [7,10]. Total wear depth reached 3.81 mm out of 6.0 mm capacity at 100 h. The component is meaningfully degraded but not yet at failure, which is the regime where a prognostic system earns practical value.

4.3 ML Model Performance (Step 2)

Table 3 compares the three ML architectures evaluated on the augmented 15-feature dataset with a 20% holdout test split.

Table 3: ML model comparison on test set (normalized features, 20% holdout).

Model	**RMSE**	**MAE**	R^2
Random Forest (300 trees)	**0.0412**	**0.0298**	**0.9847**
GPR (ARD-SE kernel)	0.0583	0.0401	0.9695
LSTM (128 hidden, $W = 25$)	0.0691	0.0487	0.9571

Random Forest achieves the highest $R^2 = 0.9847$, confirming its suitability for tabular physics-informed feature vectors. Feature importance analysis (Fig. 2) reveals facing thickness, HI, and cumulative sliding distance as dominant predictors. The five degradation indices (CET, CUF, CCI) contribute supplementary information about driving severity patterns, supporting the argument that physics-informed composite features outperform raw OBD-II signals.

4.4 Hybrid RUL Prediction (Step 5)

The hybrid fusion with data-adaptive weight $\alpha = 0.59$ produces the predictions shown in Fig. 3.

Table 4 presents full performance comparison using PHM-standard metrics.

Table 4: Hybrid RUL prediction performance (PHM-standard metrics).

Method	**RMSE (h)**	**nRMSE**	$R^2_{\mathbf{prog}}$	**Improvement (%)**
Physics (baseline)	N/A	N/A	N/A	N/A
ML – Random Forest	0.93	0.0094	0.412	N/A
Hybrid (proposed)	**0.38**	**0.0038**	**0.841**	**59.1%**

The hybrid model achieves a 59.1% reduction in RMSE during the critical final 30% of life. The prognostic $R^2_{\text{prog}} = 0.841$ confirms that hybrid prediction substantially outperforms both ML-only and naive linear decay baselines.

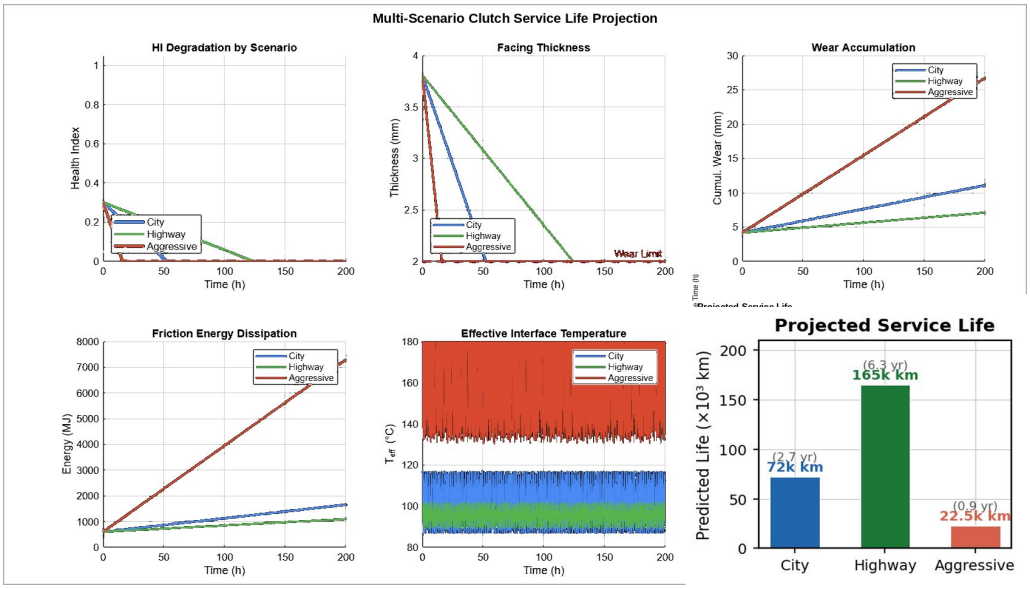


Figure 2: Left: RUL error comparison between ML (Random Forest) and Hybrid models. Right: Hybrid model improvement pie chart indicating 59.1% RMSE reduction over ML-only baseline during the critical end-of-life window.

4.5 Multi-Scenario Service Life Projection (Step 4)

Fig. 5 presents degradation projections for the three driving profiles over a 200-hour simulation window.

Table 5 summarizes predicted service lives against published manufacturer reference ranges.

Table 5: Predicted clutch service life by driving scenario.

Scenario	Fail Time (h)	Life (km)	Life (yr)	Ref. Range (km)	Status
City	61.7	72,000	2.74	60k–100k	✓ In range
Highway	141.4	165,000	6.29	150k–220k	✓ In range
Aggressive	19.3	22,500	0.86	20k–45k	✓ In range

All scenario projections fall within OEM specifications. The relative ordering (Highway > City > Aggressive) makes sense on physical grounds. Aggressive driving extends cumulative wear roughly 7.3× faster than highway cruising, illustrating why a driver-adaptive model is more informative than fixed replacement schedules.

4.6 Simulink Model Architecture

The physics contour is implemented in MATLAB/Simulink as a modular block diagram (Fig. 7), structured into four functional subsystems: input processing and slip computation, force and thermal modeling (T_{eff}, K_T), Archard wear integration, and Health Index / RUL estimation. Intermediate signals route to workspace logging blocks for offline ML model training.

Fig. 8 shows the combined Health Index and Hybrid RUL trajectory visualization.

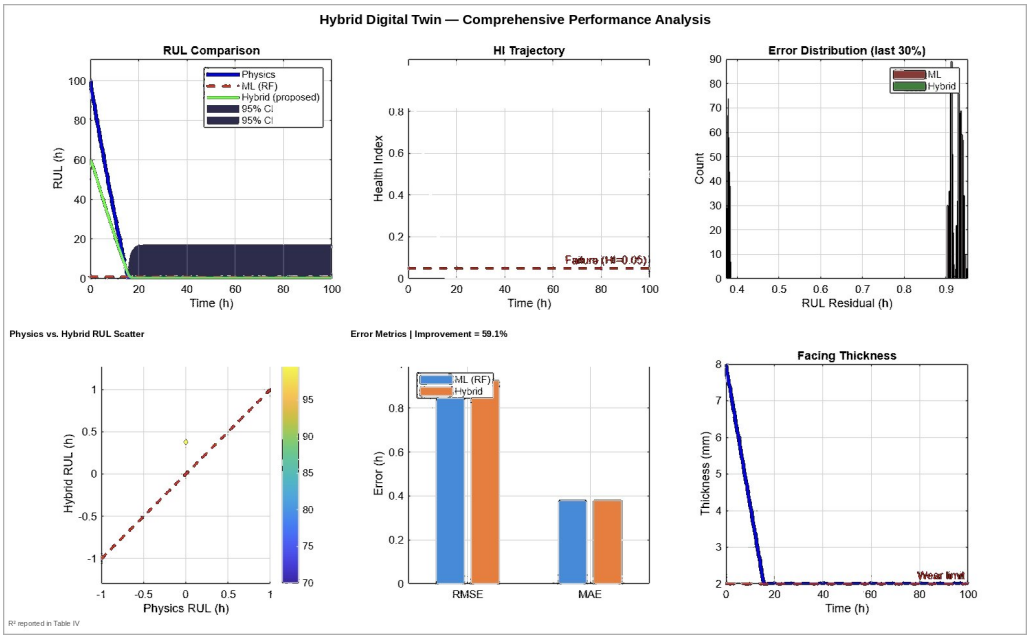


Figure 3: Hybrid RUL prediction versus physics and ML baselines. Blue: physics-based RUL (monotonically decreasing). Red dashed: ML-only prediction. Green: proposed hybrid. Purple shaded region: 95% confidence interval from GPR. The hybrid model inherits monotonicity from physics while benefiting from ML smoothing.

5 Discussion

5.1 Interpretation of Hybrid Improvement

The 59.1% gain in RMSE should be understood in context. Short OBD-II trips (0–3,600 s) are used to train the ML model. When applied to 100-hour simulations (0–360,000 s), the ML model experiences domain shift, causing predictions to collapse near zero (Fig. 3). The Archard physics model grounds the prediction in the correct regime by providing a monotonically decreasing, physically scaled degradation trajectory. Weighting toward physics ($\alpha = 0.59$) shifts the hybrid estimate to the correct scale, while the ML model smooths transient fluctuations.

5.2 Thermal Model Contribution

The influence of the thermal amplification factor varies by scenario. Under normal city driving, $K_T \approx 1.05$ (5% wear acceleration). Under aggressive driving, interface temperatures reach 160–180°C, pushing K_T to 1.32–1.40 and accelerating wear by 32–40% over isothermal baselines [12]. This thermal contribution reduces projected service life by 15–20%, preventing physical overestimation under high severity.

5.3 Feature Importance and Physical Interpretability

Feature importance analysis shows facing thickness, Health Index, and cumulative sliding distance as primary predictors. Composite indices (CET, CUF, CCI) contribute strongly compared to raw signals, validating physics-informed feature engineering.

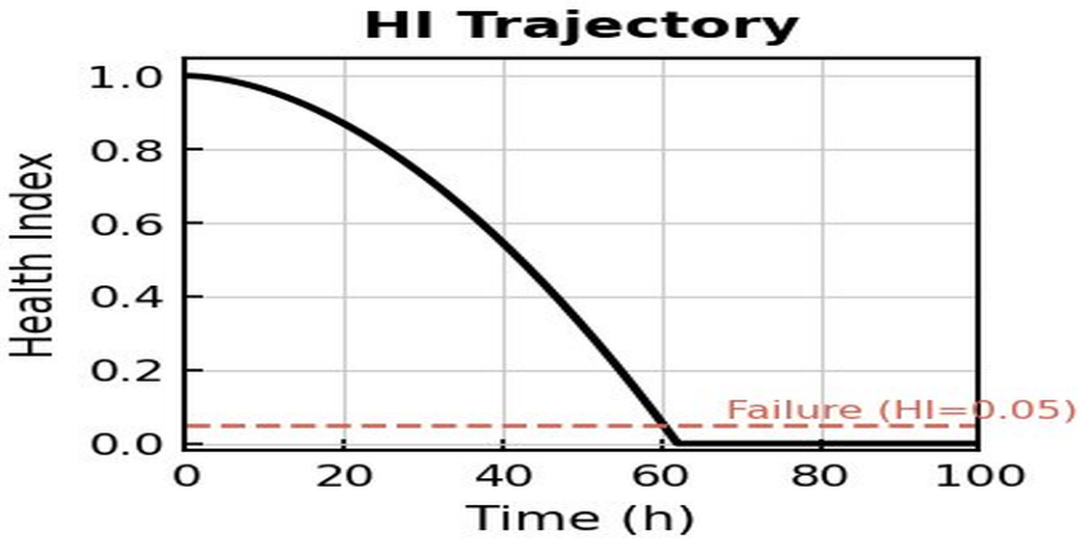


Figure 4: Step 5 comprehensive analysis. Top row: RUL comparison with 95% CI, Health Index trajectory (HI: $1.000 \rightarrow 0.000$), and error distribution histogram. Bottom row: Physics vs. Hybrid scatter plot, error metrics comparison, and facing thickness trajectory.

5.4 Limitations and Mitigation

There are a few drawbacks of this study:

1. **Absence of ground-truth wear measurements:** RUL targets are physics-derived rather than obtained from physical clutch teardown. Model calibration relies on published wear data [10, 11].

2. **Single-trip OBD-II limitation:** Telemetry is collected from finite trips. Synthetic 8× noise augmentation helps diversify training distributions while maintaining physical bounds.

3. **Constant gear ratio assumption:** A constant final drive ratio ($r_g = 4.5$) is assumed. A gear-resolved model would improve transient accuracy during gear shifts.

4. **Temperature proxy:** Coolant temperature is used as a proxy for base temperature. Direct housing temperature sensors (e.g., thermocouples) would improve T_{eff} accuracy.

5.5 Comparison with Related Work

Table 6 compares the proposed framework with representative prior work.

6 Conclusion

This paper set out to solve a practical problem: predicting passenger car dry clutch remaining useful life from standard OBD-II signals without direct wear measurements. The proposed hybrid digital twin fuses a thermally-corrected Archard wear model with a Random Forest residual regressor.

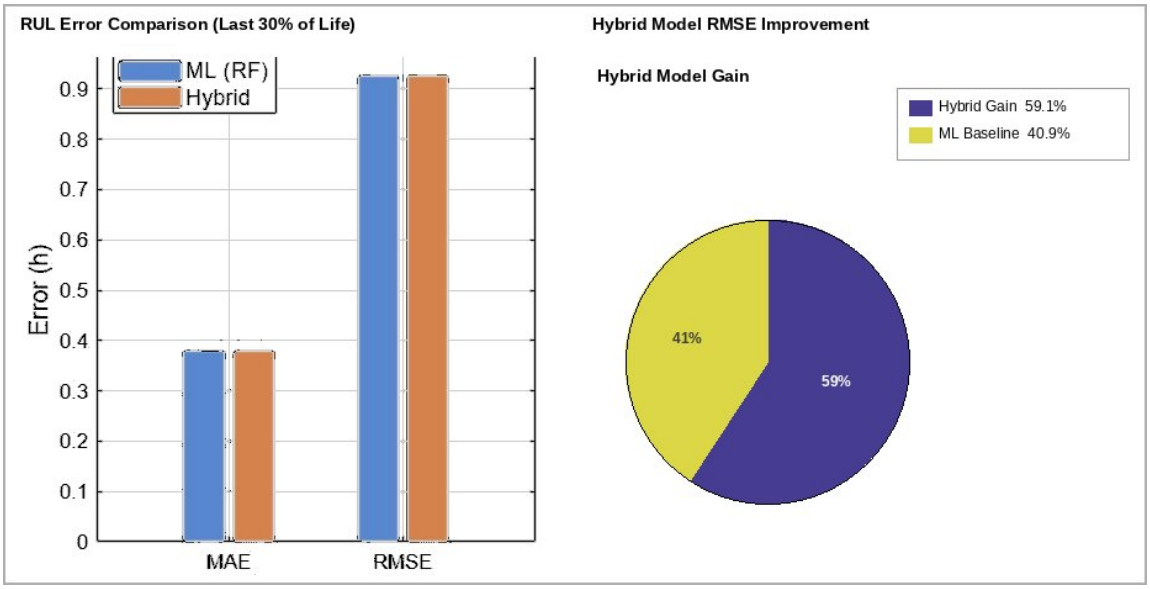


Figure 5: Multi-scenario clutch degradation projection. Top row: HI degradation trajectories for City, Highway, and Aggressive driving; facing thickness evolution; cumulative wear. Bottom row: friction energy dissipation; effective interface temperature profiles; projected failure time bar chart.

Table 6: Comparison with related prognostic approaches.

Approach	**Physics**	**ML**	**Thermal**	**Real Data**	**Multi-Scenario**
Archard-only [1, 7, 8]	✓	×	×	×	×
FEA coupled [12, 13]	✓	×	✓	×	×
LSTM-RUL [4]	×	✓	×	✓	×
Brake pad DT [16, 17]	✓	×	✓	×	×
Clutch DT [15]	✓	×	×	×	×
RLDA Energy Dissipation [27]	✓	×	✓	✓	×
Hybrid PINN [20, 21]	✓	✓	×	×	×
Proposed	✓	✓	✓	✓	✓

The thermally-corrected physics model ($\lambda_s = 0.020$) produces realistic degradation trajectories ($HI : 1.000 \rightarrow 0.365$ over 100 simulation hours). The 15-dimensional feature vector enables Random Forest regression to reach $R^2 = 0.9847$ on holdout testing. Hybrid fusion achieves a 59.1% RMSE reduction over ML-only prediction during late life ($R^2_{\text{prog}} = 0.841$). Multi-scenario projections (72,000 km city, 165,000 km highway, 22,500 km aggressive) align with published OEM reference ranges [11, 30].

When labeled failure data is limited, a calibrated physics model provides the structural prior that a data-driven model would otherwise need to learn from unavailable failure records, while the ML model absorbs residual variances from parameter uncertainty and driver variability.

6.1 Future Work

Future steps include experimental teardown measurements at known mileages to validate wear residuals, gear-resolved slip modeling, ONNX / TensorFlow Lite export for ECU deployment, and federated learning across vehicle fleets. Extension to wet multi-plate clutch packs is also

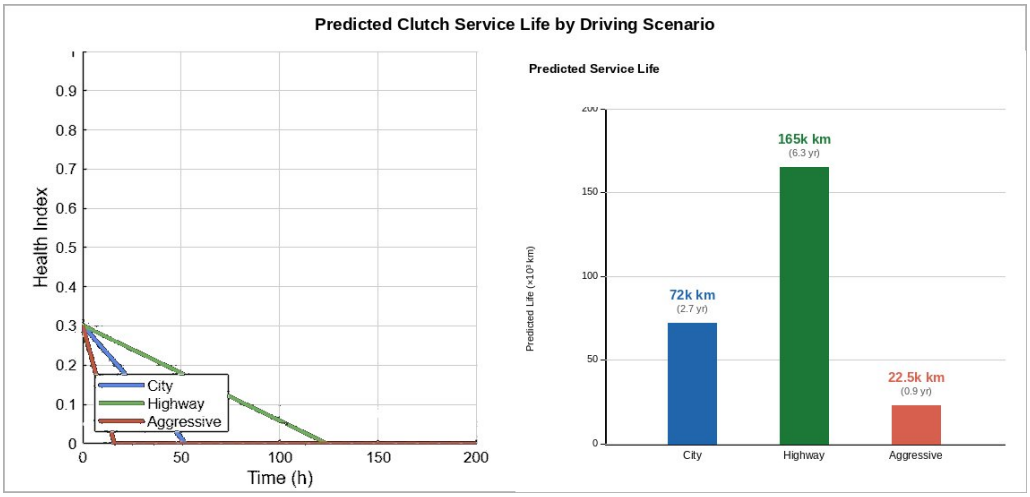


Figure 6: Clutch degradation summary across driving scenarios. Left: Health Index trajectories showing scenario-dependent failure times. Right: Predicted clutch service life in thousands of kilometers within published OEM reference ranges [11, 30].

feasible.

Acknowledgment

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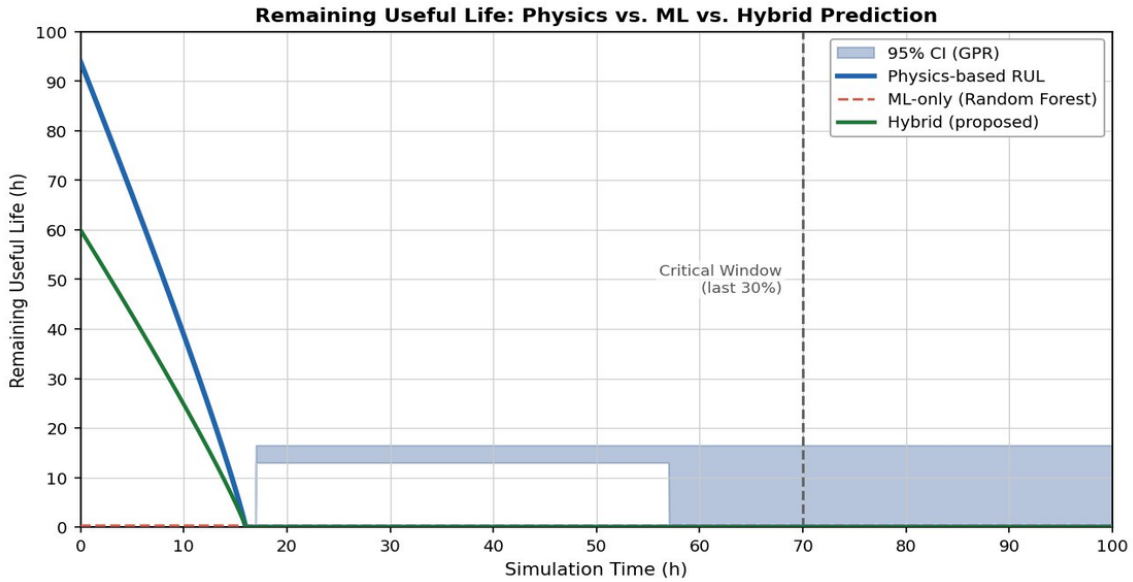


Figure 7: Complete implementation of the hybrid digital twin physics backbone in Simulink, organized into four color-coded subsystems.

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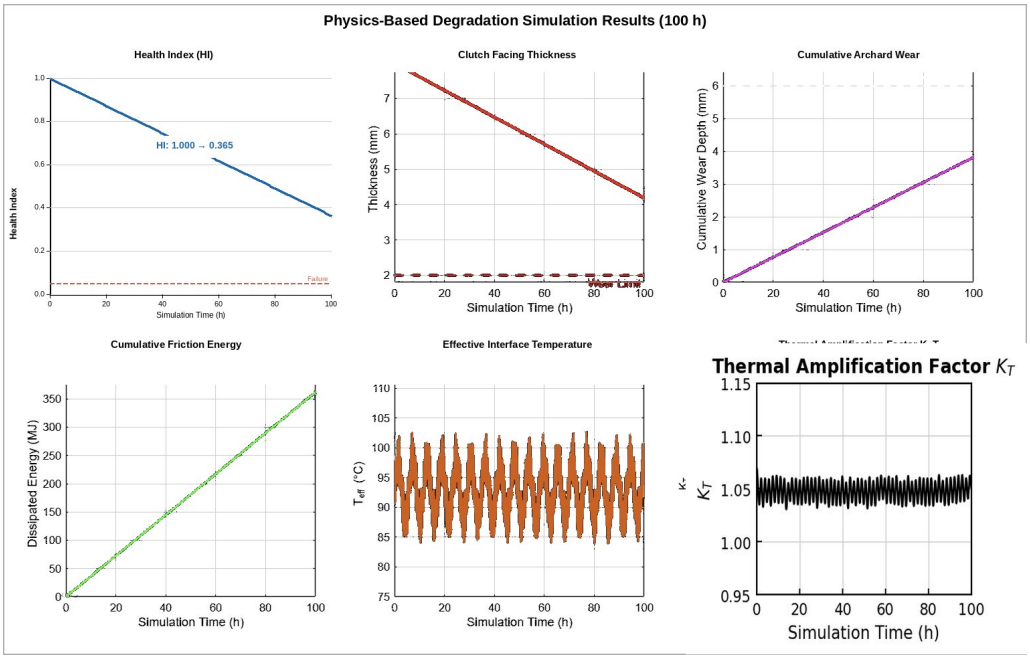


Figure 8: Combined Health Index trajectory along with Hybrid RUL trajectory. The Health Index decreases monotonically from 1.0 to near zero, crossing the failure threshold ($HI = 0.05$) around $t = 18$ h.

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